

Thank you for purchasing a FANUC Robot

READ THIS FIRST

Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation for safety procedures.

Required Tools

- ◆ Straight-head screwdriver
- ◆ Torque wrench
- ◆ Cross-head screwdriver
- ◆ Metric hex-head wrenches

Required Items

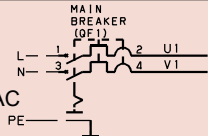
- ◆ Properly rated transport equipment as specified
- ◆ Customer-supplied properly rated power source as specified on the Power Requirements Label
- ◆ Multimeter
- ◆ Loctite #243
- ◆ Rigid platform to hold robot
- ◆ External e-stop switch
- ◆ Safety fence/gate switch
- ◆ Customer-supplied Ethernet cable
- ◆ Laptop or PC with IE, Chrome, or Safari browser
- ◆ 4-M8 bolts (SR-3iA)
- ◆ 4-M8 flat washers (SR-3iA)
- ◆ 4-M10 bolts (SR-6iA)
- ◆ 4-M10 flat washers (SR-6iA)
- ◆ 4-M16 flat washers (SR-12iA)
- ◆ 4-M16 flat washers (SR-12iA)
- ◆ FANUC-supplied USB drive cable

Optional Items

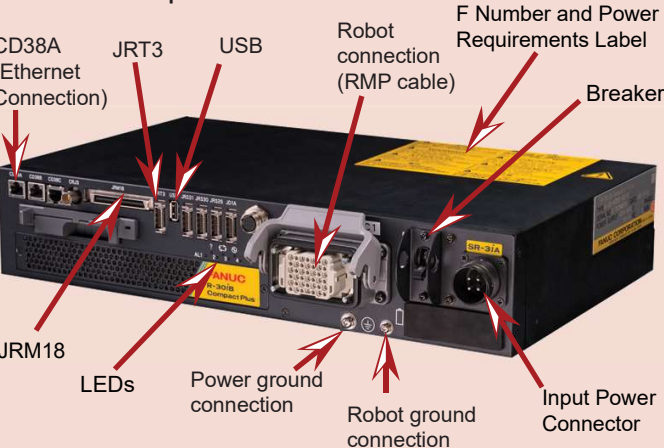
- ◆ Robot locator pins
- ◆ Auto Confirmation Signal switch

Single Phase

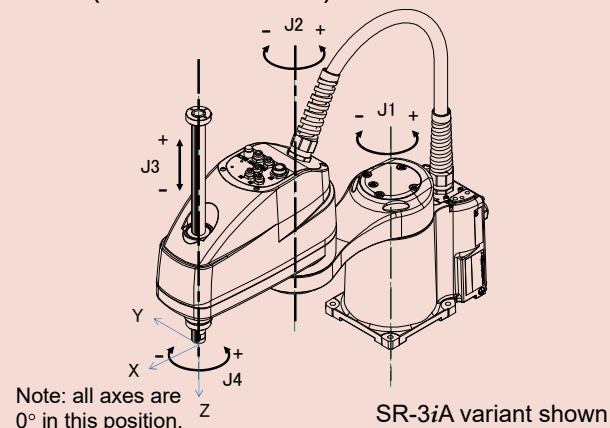
200-240VAC



R-30iB Compact Plus Controller



Robot (mechanical unit)

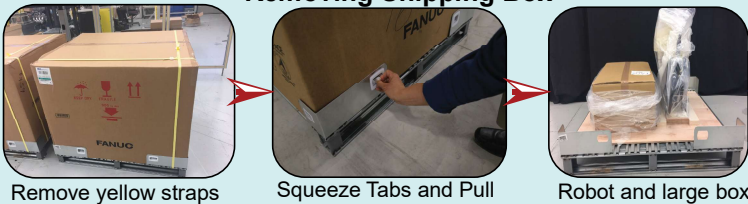


A. Unpack and Install your Robot



1. Cut the yellow shipping straps, then squeeze and pull the white tabs so the outer shipping box can be removed.

Removing Shipping Box



2. Unbox the pallet, cut any shipping straps securing the robot, and unwrap the robot.
3. Open the large box on the pallet. In the large box you will find an accessory box, your controller and, possibly, an optional Teach Pendant and Switch Box.



4. In the accessory box you will find a hardware bag (containing fuses, jumper plugs, and labels), an Inspection Data Sheet, ground cable, and a power cable.



Store the Inspection Data Sheet in a safe, known location.

5. Remove the bolts securing the robot to the transportation equipment (shown below).
6. Use properly rated equipment to move the robot to its new location. The robot alone can weigh as much as 70kg.

WARNING

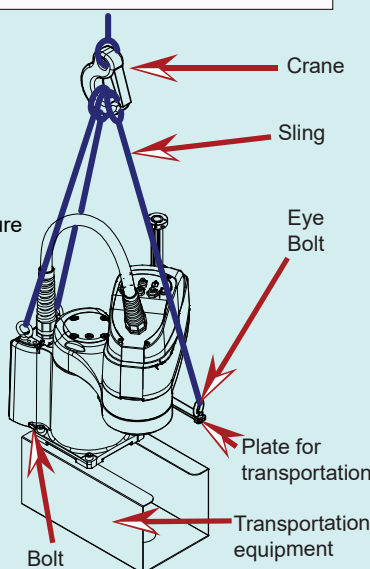
Before moving heavy equipment, check and tighten any loose bolts. Do not pull eyebolts sideways. Otherwise, you will injure personnel or damage equipment.

Crane capacity: 100kg or more
Sling capacity: 100kg or more

SR-3iA
Transport posture
J1: -20°
J2: 145°
J3: -100 mm
J4: 0°

SR-6iA
Transport posture
J1: -15°
J2: 150°
J3: -100 mm
J4: 0°

SR-12iA
Transport posture
J1: -25°
J2: 145°
J3: -180 mm
J4: 0°



7. Fasten the robot securely to the base plate (tightening torque: SR-3iA, 23-32 Nm; SR-6iA, 46-66 Nm; SR-12iA, 190-270 Nm). Use locator pins to secure the robot precisely if it is anticipated that repeated installations will be necessary in the future for service/maintenance.

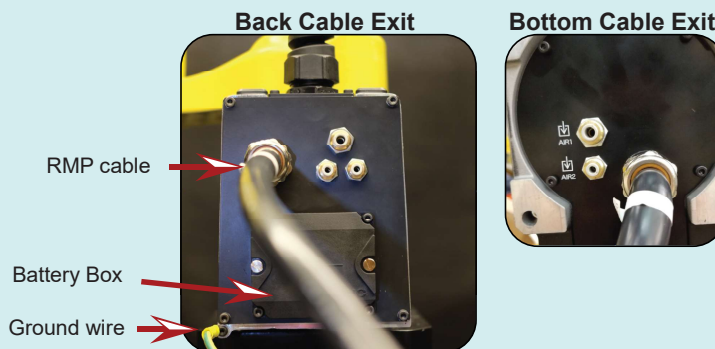
8. Remove the transportation hardware (plate for transportation, transportation equipment, eye bolts, and other hardware). Store these items and the robot skid in a safe, known location. These items are needed for future transportation.



Plate for Transportation

9. Connect the ground wire to the location on the robot base shown below.

Note: The RMP cable is attached to the robot at the factory.

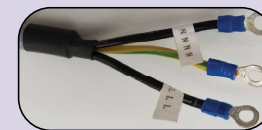


Robot installation is complete. You are now ready to go to **B. Unpack and Install your Controller**.

B. Unpack and Install your Controller

1. If you have not already done so, remove the controller from the box and place it in a safe location outside of the robot work area such that the grounding wire and robot connection cable can reach the robot.
2. Attach the RMP cable from the robot to the controller. (Refer to the controller picture to the left.)
3. Attach the other end of the ground wire that you attached to the robot in Step A9 to the controller.

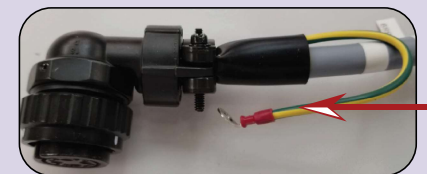
4. Have a qualified electrician connect the FANUC-supplied power cable (wires marked L, N, and ground) to an appropriate external power source.



WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

5. Connect the other end of the FANUC-supplied power cable to the Input Power Connection, located on the right-hand side of the front of the controller. Don't forget to connect the power cable ground wire (attached to the Robot Connection cable) to the power ground connection on the controller.



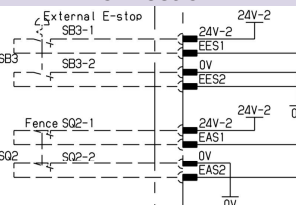
Power cable ground wire

6. Wire an E-stop switch into connector JRM18 on the controller. A partial pinout and wiring diagram showing the E-stop connections is shown below. For more detail see the *R-30iB Compact Plus CONTROLLER MAINTENANCE MANUAL*.

JRM18 Partial Pinout

	1	24VEXT_IN	35	0VEXT
	2	24V-2	36	0V
	3	24V-2	37	0V
	4	24V-2	38	0V
	5	SDICOM1	39	SDICOM2
	6	EES1	40	EES2
	7	EAS1	41	EAS2

External E-stop/Fence Connection



Note: If JRM18 is not being used, make sure the FANUC-supplied dummy plug is inserted in JRM18.

Note: If there is no Teach Pendant (TP), make sure the FANUC-supplied dummy plug is placed into the connector labeled TP.

Note: If the TP option is used without a switch box, add an "Auto Confirmation Signal" switch outside of the safety fence, wiring it to one of the available digital input signals on the JRM18 connector.

Note: If there is no switch box, make sure the FANUC-supplied dummy plug is inserted into the connector labeled JRT3.

7. Connect the customer-supplied Ethernet cable between the PC and connector CD38A on the controller.

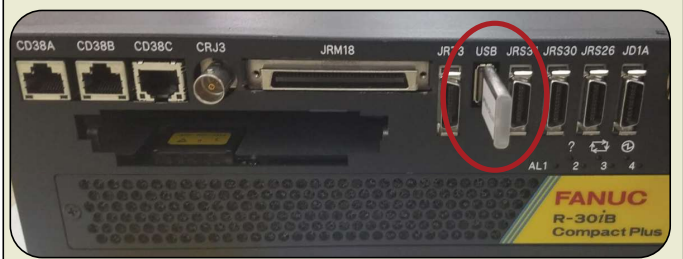
Controller installation is complete. DO NOT TURN ON THE CONTROLLER UNTIL YOU ARE INSTRUCTED TO DO SO. Go to **C. Turn on your Controller and Establish Communication**.

C. Turn on your Controller/
Establish Communication

Note: If you will be using a TP to install software, please refer to the *FANUC Version 8.X, 9.X Series Software Installation Guide*. If you do not have this Guide, please contact FANUC.

Note: If there is no need to load software (e.g. the software is the most current version available and contains all necessary options), you can proceed through the robot Initial Setup Guide and then skip to **E. Jog your Robot**.

1. Plug the FANUC-supplied USB, loaded with the robot software, into the USB port of the controller with power off.



2. Turn the Breaker ON. When the controller power-up is in progress, the Power LED will be ON and the Cycle Start LED will blink. When the Cycle Start LED stops blinking, the start up is completed.



Note: The LEDs are shown in the **Reference** section of this Startup Guide.

⚠ WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

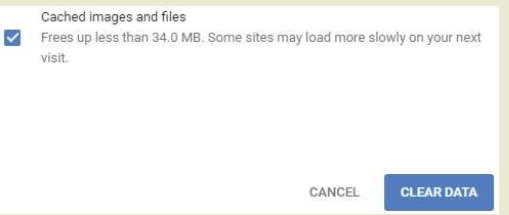
3. Connect an Ethernet cable between your laptop/PC and the robot controller.
4. Set the default address of the laptop/PC as a static IP. The PC's address should be chosen to be on the same subnet as the robot (the robot's address is 192.168.1.100). Contact your systems administrator for assistance.
5. Use a web browser (Internet Explorer (version 11 or greater), Chrome (latest version), or Safari) to connect to the robot's default IP address, 192.168.1.100. The ROBOT Homepage (similar to the one shown below) should appear. If the screen does not appear, return to the previous step and verify that the laptop/PC and robot are on the same subnet.



Note: When using IE or Chrome, make sure to clear browser cache, or screens might not display properly.

IE
Press the shortcut Ctrl-R

Chrome (shown below) Press the shortcut shift+Ctrl+delete. Select *Cached images and files* and press CLEAR DATA.

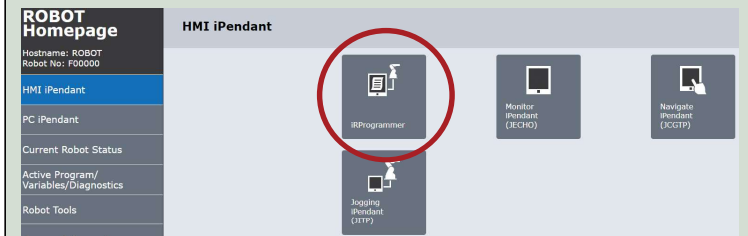


Note: If the laptop/PC will be communicating with multiple robots over a network, each robot must have a unique IP address.

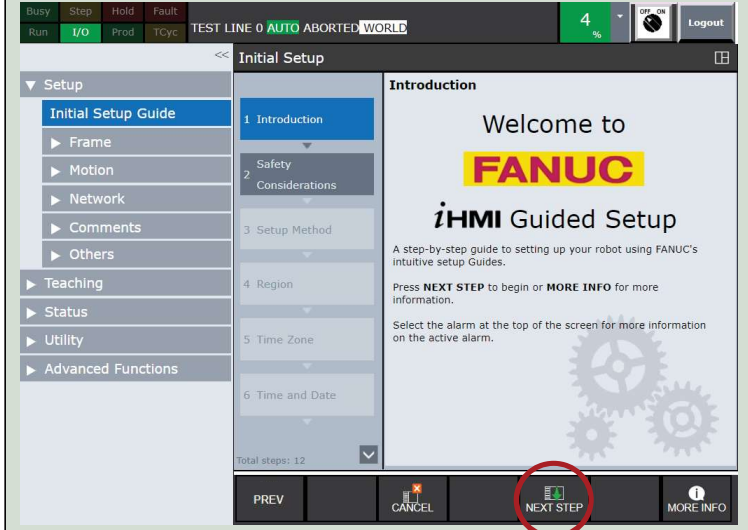
You have turned on your controller and established communication. You are now ready to go to **D. Update Software**.

D. Update Software

1. From the ROBOT Homepage, select **iProgrammer**.

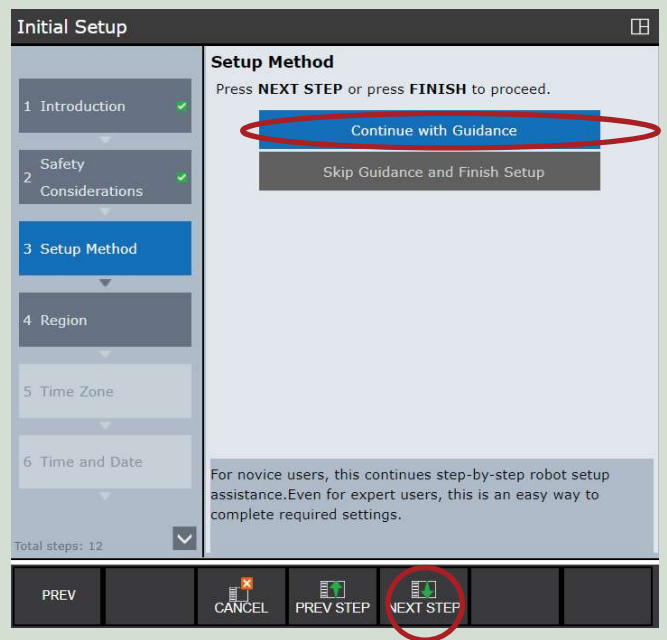


2. From the **iProgrammer** Menu on the left side of the screen, choose **Setup>>Initial Setup Guide** to display the Initial Setup Introduction screen.



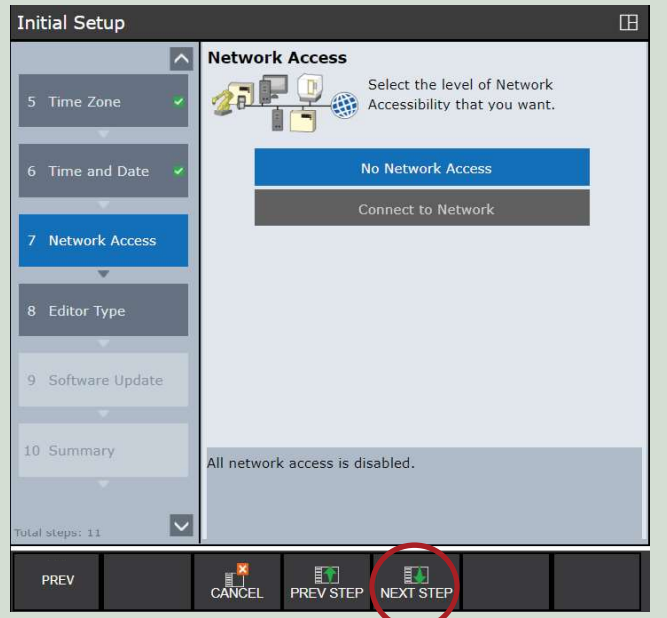
3. Once you are in the Inital Setup Introduction screen, press NEXT STEP.

4. After stepping through the Safety Considerations screen, you will see the Setup Method screen. Choose **Continue with Guidance** and press NEXT STEP.



5. Continue to press NEXT STEP, stepping through the guide and making appropriate selections, until you reach the Network Access screen shown in Step D6.

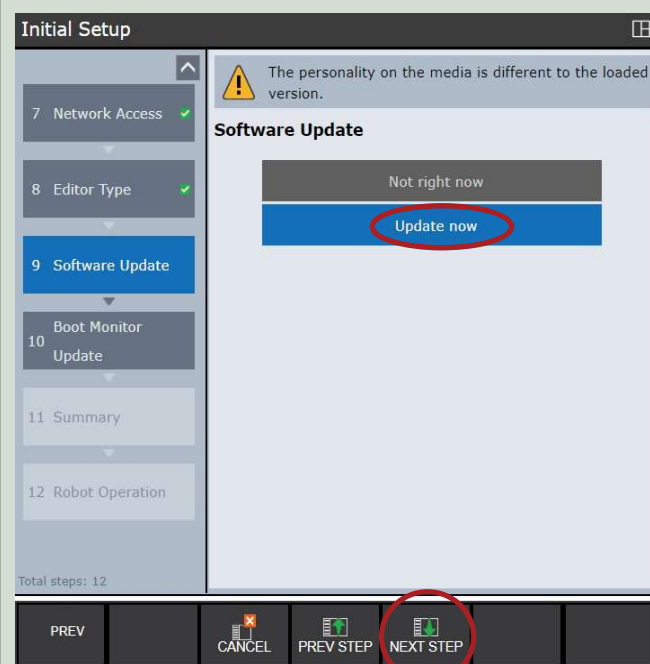
6. If you desire to change the network address of the robot from the default of 192.168.1.100, select **Connect to Network**, before pressing NEXT STEP. The Setup Guide will then guide you through this process. If you desire to use the robot's current network address, select **No Network Access**, and press NEXT STEP.



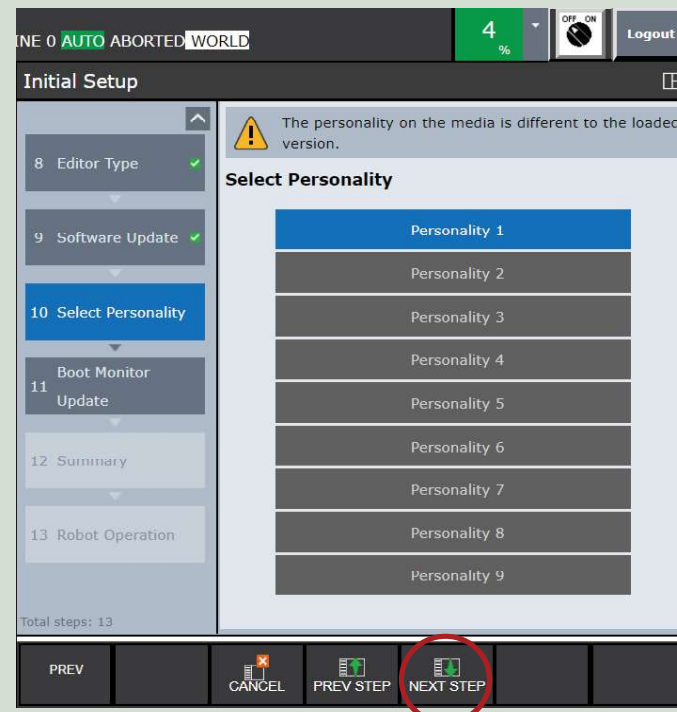
7. Continue to press NEXT STEP, stepping through the guide and making appropriate selections, until you reach the Software Update screen.

Note: If the USB contains a newer version of software, a robot personality different from the one in the controller, or multiple robot personalities, a Software Update screen will display, as shown in step 8. If the Software Update screen does not show up, continue though the Initial Setup Guide on the robot, and after finishing, skip to **E. Jog your Robot.**

- 8.** On the Software Update screen, select **Update now** and press **NEXT STEP**.



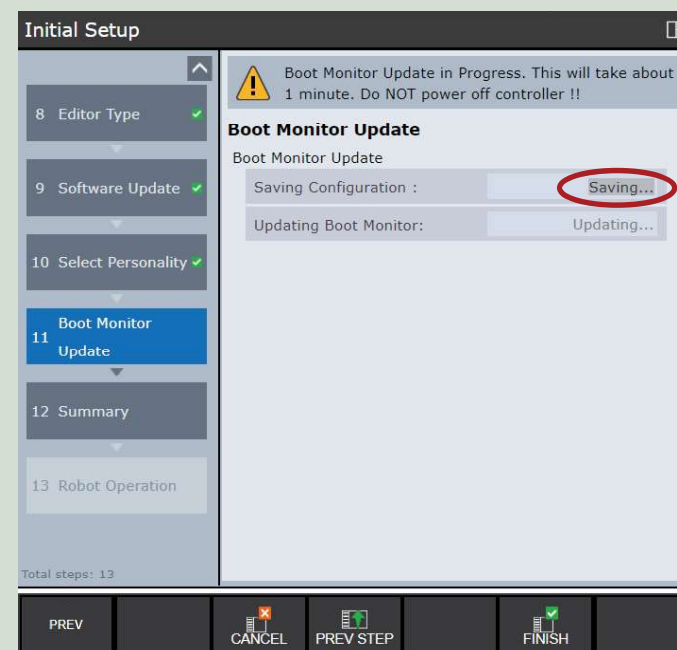
- 9.** If your USB media has more than one personality (robot configuration) on it, you will see the screen below. From the list, choose the personality you would like to load and press **NEXT STEP**. If the screen below does not appear, you do not have multiple personalities to choose from, and the controller will proceed with the Boot Monitor update.



Note: The Boot Monitor will update first. While the Boot Monitor is updating, *Saving Configuration* will indicate *Saving* as shown below. In addition, *Updating Boot Monitor* will indicate *Updating*. **During this update it is important that you not turn the robot off until the Boot Monitor has completed its update!**

⚠ WARNING

If you turn the robot off before the Boot Monitor has completed its update, serious damage can occur.

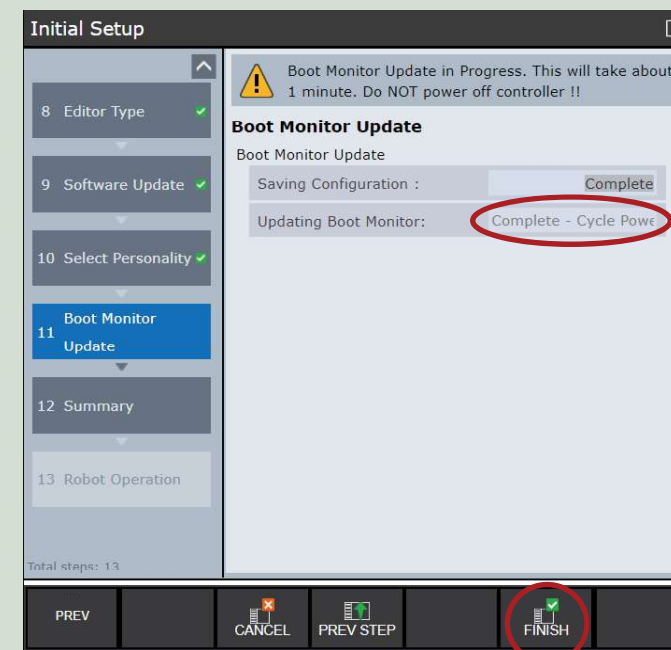


- 10.** Once the Boot Monitor Update is complete, an error message might appear reminding you to cycle power. Once the error message appears, proceed to step 12. If there are other existing faults, the error message might not appear. In this case, proceed to step 11.



- 11.** If the **Please cycle power** message does not appear but the Boot Monitor Update is complete, as indicated in the screen shown below, the FINISH button will appear. If this occurs, press FINISH and the **Please cycle power** message will display.

Note: Make sure that the Boot Monitor update is complete as indicated below. **If you turn the robot off before the Boot Monitor has completed its update, serious damage can occur.**



- 12.** Turn the controller off and then back on again.

Note: All previous setup will be saved by the system prior to the software update, and automatically restored after the update.

- 13.** Cycling power will terminate communication between the robot controller and the Laptop/PC. Therefore, on the Laptop/PC, it is recommended that you close the iRProgrammer window after cycling robot power.

- 14.** About 10 seconds after turning on power, the Cycle Start LED will start blinking. The Cycle Start LED will continue to blink for about 15 minutes, after which it should stop blinking, indicating that the software has finished loading. During this time, do not attempt to remove the USB or turn off the controller.

⚠ WARNING
While the robot is powering up, do not attempt to remove the USB or turn off the controller.

You have updated your software. You are now ready to go to **E. Jog your Robot.**

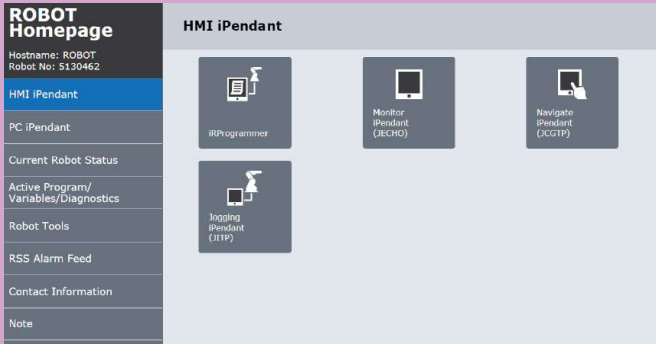
E. Jog your Robot

⚠ WARNING

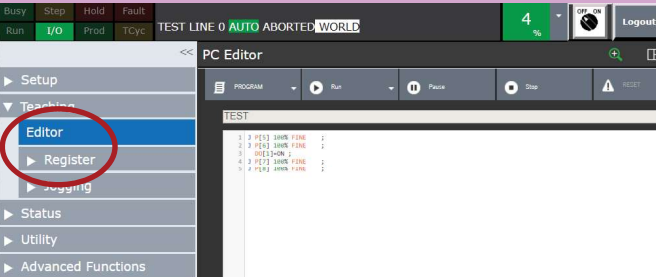
When you use the jog panel to execute a jog operation, observe the following:

- The jog operation must be executed from outside the safeguards.
- Make sure that no one is inside the safety fence.
- An operational Emergency Stop button must be near the operator.
- Jogging must be executed in the sight of the robot.

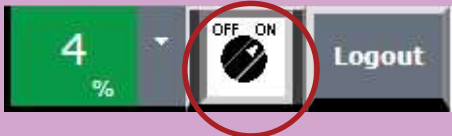
1. Use a web browser (IE, Chrome, or Safari) to connect to the robot's default IP address, 192.168.1.100. The ROBOT Homepage shown below should appear.



2. Make sure HMI iPendant is selected on the left.
3. From the ROBOT Homepage, select iRProgrammer.
4. Select Teaching, and choose Editor.



5. Turn the TP enable key to ON.



6. You might have to reset a fault. If so, press the RESET button at the top of the screen.



Note: You might see an alarm indicating that DCS has changed. If so, go to **F. Apply DCS Params** and follow the steps. Afterwards, return to Step 7 in this section to continue.

7. The lower right side of the iRProgrammer PC Editor page (see **Reference** in this guide) has a jog panel that can be used to jog the robot. To move each joint axis of the robot, press each joint jog key (labeled J1 - J4) one at a time and watch the robot joint move. Refer to the Robot (mechanical unit) with joints indicated diagram on Page 1 of this guide.

8. If you wish to adjust the jogging speed, press the arrow next to the current speed (in this case 4%), and use the slider bar to adjust the speed up or down.

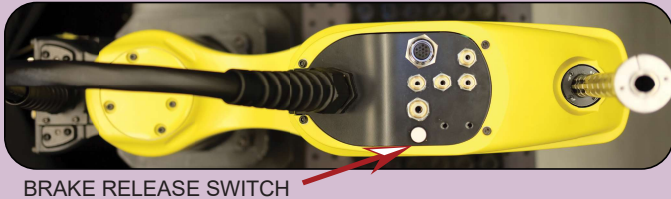


Note: In addition to jogging the robot, you can also move the robot by hand-dragging each axis.

CAUTION: If it is desired to move axis 4 by hand, the brake release switch must first be pressed. Be aware that, with this button pressed, gravity may cause axis 4 to move unexpectedly.

CAUTION: If a person is inside the perimeter guarding to move the robot by hand, the power to the servo motors must be cut-off and locked out to prevent automatic operation.

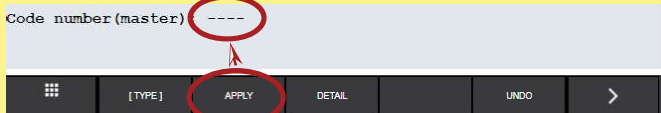
9. To hand-drag the robot, make sure the TP enable key is off and the **fault has not been reset**. Then physically grab and move the robot.



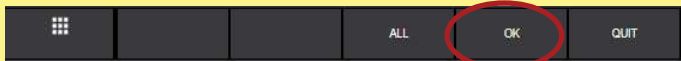
You have installed your robot.
You are now ready to add tooling and program your application

F. Apply DCS Params

1. To get to the DCS screen, from within the iRProgrammer Menu, select **Setup>>Others>>DCS**.
2. The screen that you see will indicate that there has been a change that affects DCS. Press APPLY to apply the change to DCS, and enter the Code number (default is 1111).

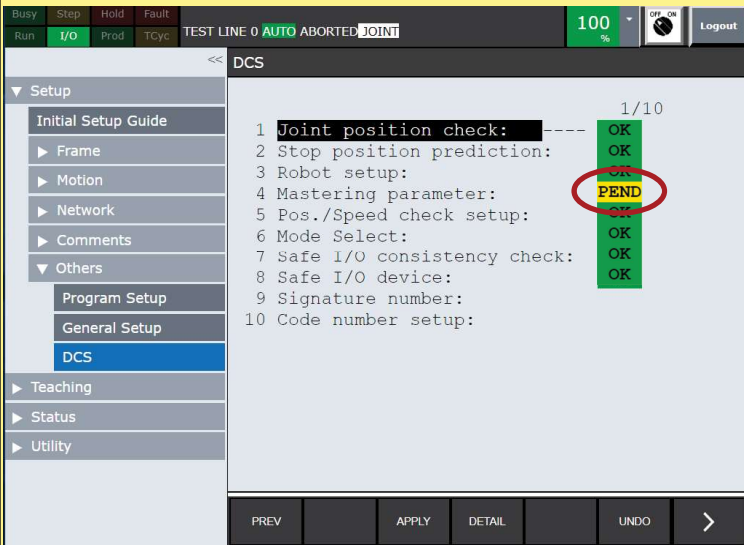


3. Press OK.



4. The changed parameter should now indicate PEND. Cycle power to apply the DCS change.

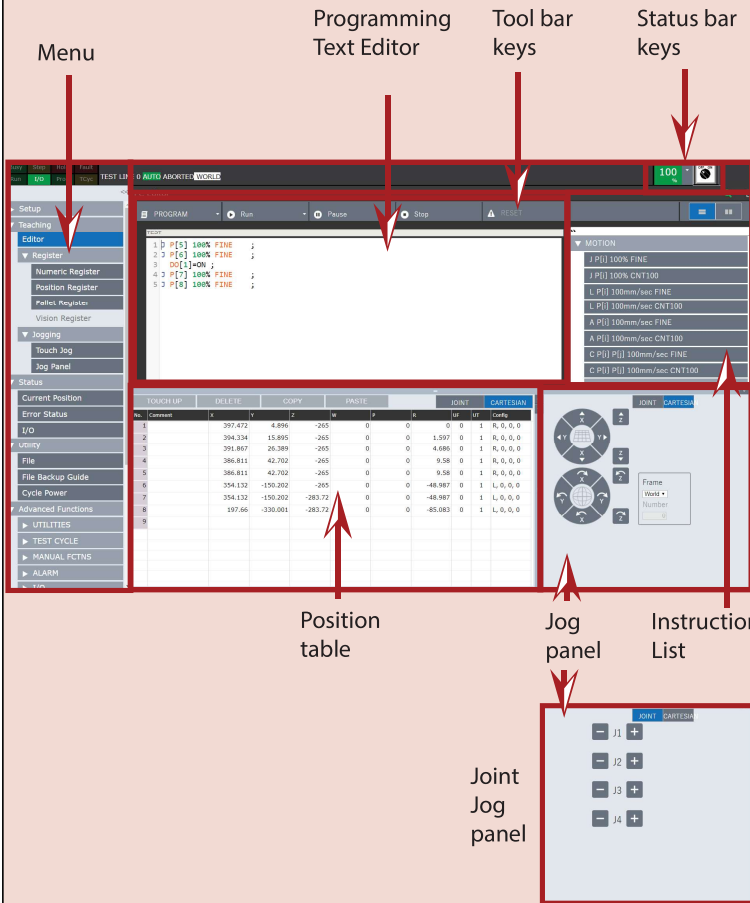
Note: There might be more than one changed parameter.



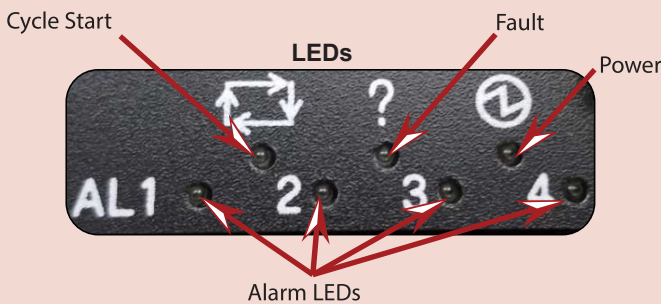
5. If you left **E. Jog your Robot**, Step 6, in order to Apply DCS Params, you may now return to **E. Jog your Robot**, Step E7 and continue.

Reference

iRProgrammer™ PC Editor screen



LEDs



Note: For the purpose of this guide, the Alarm LEDs are not used.

For more information

To access controller, robot, or application-specific manuals, scan the QR Code to the right or go to <https://www.fanucamerica.com/qr/edocs>.



United States: 1-888-326-8287

Canada: 1-905-812-2300

Mexico: + 52 (449) 922-8000

South America: + 55 (11) 3619-0599

Or, select your region at <https://www.fanuc.com>.